lipid extract was layered with 1/ 5th volume of normal saline (0.9% NaCl) and mixed several times by gentle inversion. It was allowed to stand for 6 to 8 hours at room temperature. The lower chloroform layer was collected, evaporated to dryness in sand bath at 45°C-50°C and was repeated thrice. The extract was dissolved in known volume of chloroform. The total lipid extract were determined aluminium/stainless steel planchets and dried at 40°C - 50°C in an oven until constant weight were reached. Increased weight gives total serum lipid. Estimation of Phospholipids. The total phospholipid in the serum was estimated by method of Post and Sen (1967)[3]. An aliquot 0.1 of lipid extract was taken into micro kjeldahl flask and 0.4 ml of 60% perchlororic acid solution was added. It was digested directly on a sand bath for 20 minutes. Glass beads (2-3) were added to each flask to avoid bumping during digestion. The 0.1 ml of digested solution was taken in a tube and was added to 8 ml water. It was then mixed with 2.0 ml coloring reagent (1 part of 10% ascorbic acid and 6 part of 0.42%) was added. It was incubated in heated water bath at 37°C for 1 hour, after mixing absorbance was taken at 660 nm in colorimeter AE model Japan against H_2O. Which was expressed as Mg lecithin/ml blood data collected, different traits were analyzed as per the standard statistic techniques.[4-6]

RESULTS & DISCUSSION

Total serum protein content of treatment and control group of Rabbits are presented in Table 1. The analysis of variance of this data in presented in Table - 2. The result showed significant difference ($P < 0.01$) in total serum protein content between treatment and control group. The significantly higher value in treatment group was observed but non-significant differ between sexes were find such differences in protein content of serum was clearly evidenced.

The value of total blood glucose level of control and treatment groups of rabbits are summarized in Table - 3. The analysis of variance of this data is presented in Table-4. The finding indicated significant ($P < 0.01$) difference between groups. But no any changes between sexes were observed.

The total serum cholesterol levels of different groups of rabbit are summarized in Table-5. The analysis of variance of this data presented in Table - 6. The finding showed significant differences ($P < 0.01$) in total serum cholesterol level among treatment and control group and non significant between sexes. However increased trend was observed.

The total lipid level in serum of treatment and control groups are presented in Table - 7. The analysis of variance of this data presented in Table - 8. The presented findings showed no significant difference in total lipid level between control and treatment group. But significant ($P < 0.01$) difference found between male and female. Male showed higher value than female, such observation are clearly visible.

The total phospholipid content in serum is presented in Table 9. The analysis of variance of this data presented in Table - 10. The present findings showed non-significant difference between control and treatment group also did not showed difference between sexes. The decreased levels of total phospholipd are clearly visible.

Table-1: Showing the values* of total serum protein (gm %) of different groups of rabbit.

Group No.	Sex	Total serum protein	
		Sex	Group
Control	Male	5.44±0.134	5.15±0.184
	Female	4.82±0.167	
Treatment	Male	6.58±0.134	6.41±0.106
	Female	6.24±0.167	

* Value (Mean ± SE) bearing the same superscript did not differ significantly.

Table-2: Analysis of variance of the data presented in table 1 showing the effect of treatment and sex on total serum protein content in rabbit.

Source of Variation	d.f.	S.S.	M.S.S.	F
Between group	1	4.78	4.78	29.87**
Between Sex	1	0.75	0.75	4.7^{NS}
Error	9	1.45	0.16	

**- Significantly at 1% level ($P < 0.01$)
NS- Non significant

Table-3: Showing the values* of total blood glucose (mg %) of different groups of rabbit.

Group No.	Sex	Total serum protein	
		Sex	Group
Control	Male	139.3±7.86	142.38±10.89
	Female	145.4±4.24	
Treatment	Male	206.03±7.86	193.91±7.65
	Female	181.40±4.24s	

*Value (Mean ± SE) bearing the same superscript did not differ significantly.

Table-4: Analysis of variance of the data presented in table 3 showing the effect of treatment and sex on total glucose content in rabbit.

Source of Variation	d.f.	S.S.	M.S.S.	F
Between group	1	7967.05	7967.05	11.78**
Between Sex	1	248.43	248.43	0.36^{NS}
Error	9	6085.19	676.13	

**- Significantly at 1% level (P < 0.01)
NS- Non significant

Table-5: Showing the values* of total serum cholesterol (gm %) of different groups of rabbit.

Group No.	Sex	Total serum protein	
		Sex	Group
Control	Male	48.50±1.96	46.08±4.54
	Female	43.83±1.23	
Treatment	Male	65.16±1.96	59.75±2.6
	Female	54.33±1.23	

* Value (Mean ± SE) bearing the same superscript did not differ significantly.

Table-6: Analysis of variance of the data presented in table 5 showing the effect of treatment and sex on total serum cholesterol content in rabbit.

Source of Variation	d.f.	S.S.	M.S.S.	F
Between group	1	572.28	572.28	6.81**
Between Sex	1	233.12	233.12	2.77^{NS}
Error	9	755.51	83.94	

**- Significantly at 1% level (P < 0.01)
NS- Non significant

The total protein content of honey treated rabbits was found to be increased significantly as compared to control, suggesting anabolic effect of honey in rabbit. However no effect on sex was observed.

There is highly significant effect of honey treatment is event as increased level of blood glucose was found which provide energy and non-significant effect on sex was observed.

There is significant effect of honey on total cholesterol content among group's increases but non-significant effect on sex was observed. In total lipid content show non-significant effects on group but have significant effect on sex. There is non-significant effect of honey on serum phospholipid among group and sex. The anabolic effect of honey evident from present result is on protein, lipid and carbohydrate metabolism.

There are significant differences in the level of lactate dehydrogenase and succinic dehydrogenase among groups. LDH shows significant and SDH shows non-significant effect on sex, these observations indicating variation on glycolytic pathway. On the other hand effect of treatment as found on Kreb's cycle which provide the major source of energy to the animals.

The effect of treatment on ALT & AST were non-significant indicating no influence on protein atabolism only ALT value slightly increase between groups.

The effect of treatment evident on ATPase activity as well as alkaline phosphatase such inding suggested significant effect of honey on energy transfer process in rabbit. There is no significant differences were observed in biochemical haemoglobin polymorphism which suggested further details studies using different breeds.

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Change in the fundamental niche of *Mesocyclops* species by the action of *Brachionus* species at Ranchi Lake

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Abstract- Changes in the niche structure of a cultural eutrophic lake at Ranchi were studied in relation to the population structure of three zooplankters *Mesocyclops leuckartii, Brachionus qudridentatus and Brachionus angularis*. Twenty two parameters were used to define the broad environmental framework of the lake. These parameters create an n-dimensional hypervolume within which the plankton selects the optimum combination of variables that creates its fundamental niche. The fundamental niche of *Mesocyclops* is defined by three parameters: food resource, temperature, and alkalinity. However, the same food resource is used by other zooplankton and thus compresses the fundamental niche to a realized niche. The parameters defining the fundamental niche are in turn are controlled by a variety of other subsystems. The result showed that the *Brachionus* species caused a change in the fundamental niche of *Mesocyclops* species.

Key words: Hypervolume, temperature, alkalinity, niche, correlation

INTRODUCTION

Rotifers of the Subclass Monogononta (Phylum Rotifera) are common metazoans in the plankton of freshwater environment.[1,2] Although small (usually 50–500 µm), they often are extremely abundant and important in the transfer of energy from algae, microbes, and detritus to higher trophic levels in community food webs.[3]

Rotifer life cycles often include several types of females and eggs.[4] Females sequentially produce single eggs throughout their reproductive life. Most rotifers are oviparous for instance in *Brachionus* eggs are attached to the mother until they hatch[5] and are released into the environment, or deposited onto a substratum.

An early, long – held view of the rotifer life cycle was a seasonal progression of events in which resting eggs hatch, populations develop via female parthenogenesis as long as environmental conditions are favourable, and then bisexual reproduction with resting-egg production occurs when conditions turn unfavourable at the end of the growing season.

The objective of the present study was to identify the changes in the niche structure of *Mesocyclops* species which shrinks its fundamental niche to a realised niche by the action of another species *Brachionus* at Ranchi Lake, Ranchi.

MATERIALS & METHODS

Plankton samples were collected once each month and used for species identification and counting on a Sedgwick rafter cell at 10x magnification. Twenty two parameters were used to define the broad environmental framework of the lake.[6] An n-dimensional hyper volume is created by these parameters. The plankton selects the

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optimum combination of variables among these which create its fundamental niche. The estimation of chlorophyll was done by using spectrophotometer at 630nm, 645nm and 665nm.[7] The measurement of water temperature, salinity, conductivity and alaklinity[8] was done every month.

RESULTS & DISCUSSION

Results are presented in Table 1 and Figures 1 to 9. Figure 1 (a) and 1 (b) depict the *Brachionus* species with attached eggs. Bisexual reproduction is clearly evident in *Brachionus* species (Figure 2).

At Ranchi Lake three zooplankters were in abundance that is *Mesocyclops leuckartii, Brachionus qudridentatus and Brachionus angularis* (Figure 3). Organisms are primarily dependent on their food source. *Mesocyclops* is omnivorous feeding on algae, insect larvae and dead organic matter.[9,10] However, their primary dependence is on available algae.

Figure 4 represents changes in the phytoplankton population with respect to food availability at Ranchi Lake during the study. Another aspect which affects the plankton is competition from *Brachionus* which in the present case is *Brachionus angularis*. Populations of *Brachionus angularis* were quite common in Ranchi lake and had a fundamental niche defined by a temperature of 15-25ºC; a salinity of 9-40 (parts per thousand %); conductivity of 5mS cm^{-1} and food resource. The average life span of *Brachionus* species is 8-10 days and produces two eggs at a time. The maximum individuals were found in October at a temperature of 24.7°C and minimum in March at a temperature of 26.4°C (Figure 4).

The basic energy source of the zooplankton is the phytoplankton population specifically the various species of algae. The phytoplankton population at Ranchi Lake showed two peaks, one in the month of April – May, and the other in the month of August – September (Figure 5), it exhibited fluctuations with nutrient availability as the nutrients were released through decomposition of organic matter (Figure 6).

The phytoplankton population reached its peak in April-May and decreased rapidly in June due to unfavorable environmental conditions. It was estimated by chlorophyll method. The phytoplankton population was calculated from the concentration of chlorophyll and a significant negative correlation (r = -0.98) was noted between the organic matter and the phytoplankton population (Figure 7).

Brachionus competes for food resource to a certain extent but it has an alternative food source in the form of dead particulate organic matter. However, *Mesocyclops* can change its food preference from, algae to predation on living zooplankton, including *Brachionus* and dead organic matter.

One of the competitors of *Mesocyclops* with respect to food resource is *Brachionus*. A significant negative correlation (r = - 0.82) was noted between the population of *Mesocyclops* and *Brachionus* (Figure 8). *Brachionus* species shrinks the fundamental niche of *Mesocyclops* species to a realized niche (Figure 9).

$\Delta \log$ Mesocyclops = - 0.5481Δ log Brachionus – 0.0278 (r = - 0.82)

Brachionus uses the same food resource as *Mesocyclops* and competes with the latter for food. This competition shrinks the fundamental niche of *Mesocyclops* species to a realized niche (Figure 9 & 10).

Table 1: The parameters and their range

Parameters	Range
Temperature	15 – 25 °C
Salinity	9 – 40 ‰
Conductivity	5 mS cm^{-1}
Alkalinity	200 – 250 mgL^{-1}
Chlorophyll	12.55 μgL^{-1}

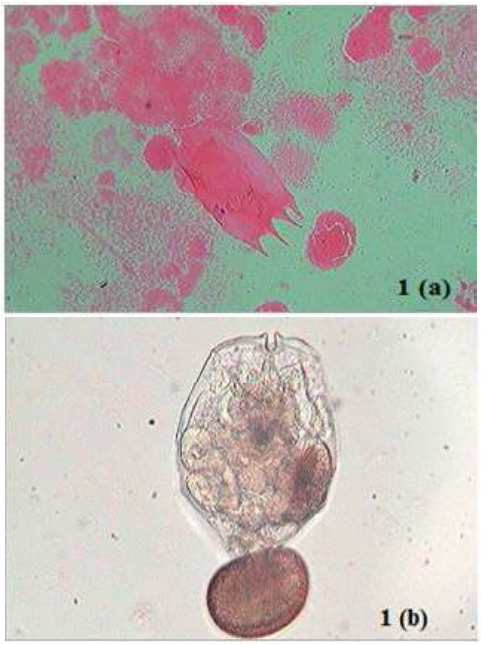


Fig 1: Eggs are attached to the female of *Brachionus angularis*